

Boundary-centered model of public laboratory-automation bottlenecks

Ecosystem knowledge and support

B1 Knowledge packaging B10 Documentation, training, support



Interfaces and representations

B2 Device access B3 Deployment identity B4 Physical resources



Runtime coordination

B5 Observability B6 Recovery B7 Scheduling



Evaluation

B8 Test-claim alignment

Emerging AI

B9 Context and physical feedback

The model organizes analytical constructs; it does not claim exhaustiveness or population prevalence.